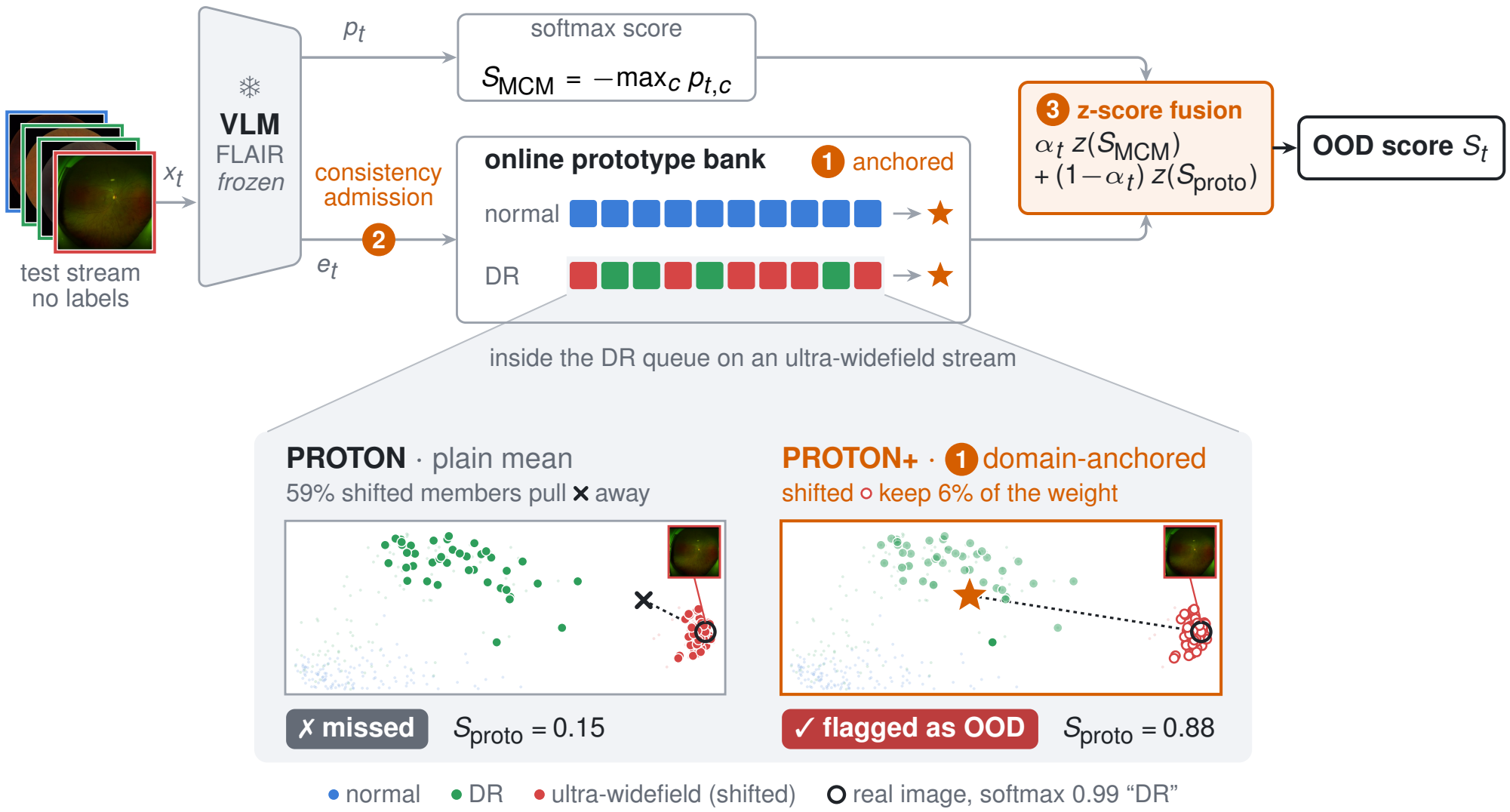

test stream
no labels
x_t
VLM
FLAIR
frozen
p_t
softmax score
$S_{MCM} = -\max_c p_{t,c}$
consistency
admission
2
e_t
online prototype bank
1 anchored
normal
DR
3 z-score fusion
$\alpha_t\, z(S_{MCM})$
$+ (1-\alpha_t)\, z(S_{proto})$
OOD score S_t
inside the DR queue on an ultra-widefield stream
PROTON · plain mean
59% shifted members pull ✕ away
✗ missed
$S_{proto} = 0.15$
PROTON+ · 1 domain-anchored
shifted ○ keep 6% of the weight
✓ flagged as OOD
$S_{proto} = 0.88$
normal
DR
ultra-widefield (shifted)
real image, softmax 0.99 “DR”